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ITS66504: Systems Fundamentals

ASSIGNMENT 2

COVER PAGE

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Instructions to student:

- This is a **GROUP** assignment.
- Answers **MUST** be in PDF format.
- Submission due date is on the **28 DECEMBER 2024 11.59PM** via myTlmeS.

Student declaration:

I declare that:

- *I understand what is meant by plagiarism*
- *The implication of plagiarism has been explained to me by our lecturer*
- *This assignment is our own work*

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1.0 Relationship between Moore's Law and Virtualization

Gordon Moore, the co-founder of Intel, predicted in 1975 that the number of components on a single chip would double every year. His prediction was based on observing a trend where the number of transistors on an integrated circuit at minimum cost had doubled between 1960 and 1965 (Tardi, 2024b). This observation is known as Moore's Law. This law suggests that, as time progresses, computers, devices powered by computers and computing capabilities become smaller, efficient and more affordable as processes improve and components advance (Tardi, 2024b).

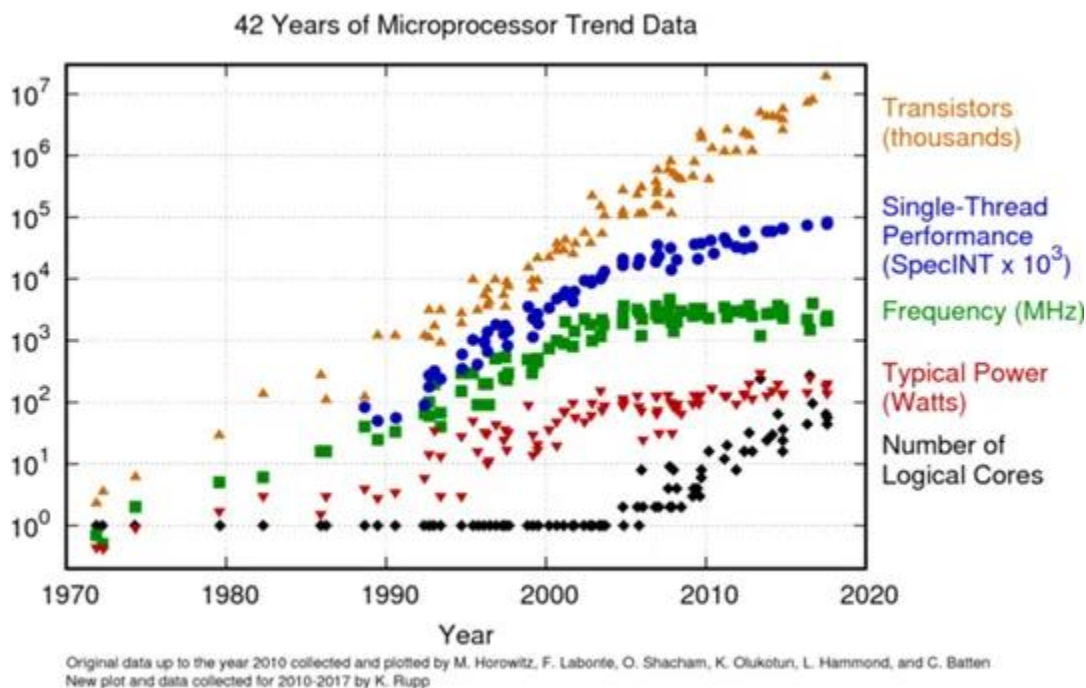


Figure 1-Moore's Law Chart

Before the 21st century, CPU clock speeds were rapidly increasing in months due to the shrinking of transistors size which allowed more transistors to fit in a single chip. Furthermore, reducing the voltage used by these transistors further improved the speed. However, by the early 21st century, the ability to lower the voltage has reached its limits, despite this, reducing the size of transistors could still be done which led manufacturers to add multiple CPU cores to processors. This change transformed the industry as processors with multiple cores made it possible to run several virtual machines on a single

physical server. Although the development pace of CPU might not be as fast as it used to, this shift allowed virtualization to be directly supported in hardware, enabling more efficient computing systems (Lyons, 2021).

As Moore's Law slows down and the demand for big-data applications grows, virtualization has become more important (McClain, 2020). Although there are limited upgrades available to work on for the CPUs, virtualization will not end; it will adapt to the new challenges. Advancements on components such as storage and networking are essential to compensate for the slow progress in CPU development. Virtualization involves the allocation of memory however, as datasets grow and computer memory is scarce, managing memory will be challenging, the need for efficient allocation of memory is needed (McClain, 2020). By improving system architectures other than CPUs, virtualization can continue to thrive, even if Moore's Law ends.

1.1 Server Virtualization

Server virtualization is the process of separating a physical server into multiple isolated virtual servers using software, allowing them to run its own operating system and applications independently (NUTANIX, 2023). This helps improve the processor's utilization by allowing the servers to operate at full capacity, preventing idle resources as traditionally, workloads are only handled by a small portion of the servers (ABACUS, no date). It also reduces the need for additional hardware (ABACUS, no date). As a result, it saves physical space, reduces overcrowding in data centres and lowers energy consumption therefore, lowering business' overhead. Additionally, server virtualization hides server resources such as the number and identity of physical servers, processors, and the various operating systems (Bigelow and Gillis, 2024).

How server virtualization works by adding a layer of software, known as a hypervisor; it acts as a controller which manages and allocates resources like processors, memory, storage and networking into independent logical instances called virtual machines (VMs) (Bigelow and Gillis, 2024). The hypervisor can dynamically reallocate resources between

VMs if needed, ensuring optimal and efficient server resource utilization (NUTANIX, 2023). There are three types of server virtualizations: Full Virtualization, Paravirtualization and OS-level Virtualization.

1.2 Full virtualization

Full virtualization is the most used virtualization type. This type uses a hypervisor to directly communicate with the computer's hardware and run different operating systems in one server independently (ABACUS, no date). It has two types of full virtualizations which are Software-Assisted Full Virtualization and Hardware-Assisted Full Virtualization.

Software-Assisted Full Virtualization translates instruction sets using binary translation for the system to understand, allowing it to manage and virtualize tasks (ABACUS, no date). It uses software to replicate hardware for virtualization to happen. Examples of this kind of virtualization are:

- Virtual PC
- VMware workstation (32-bit guests)
- VirtualBox (32-bit guests)
- VMWare Server

Hardware-Assisted Full Virtualization directly uses the computer's hardware depending on the instructions which can be executed on the processor (ABACUS, no date). It uses two types of hypervisors which are Type 1 Hypervisor and Type 2 Hypervisor. Type 1 Hypervisor, or the bare-metal hypervisor type, is installed directly on top of the underlying machine hardware (AWS, no date). It allocates resources based on the hardware and once used, it cannot be used for anything other than virtualization (ABACUS, no date). On the other hand, Type 2 Hypervisors are installed within the host computer's operating systems. They are widely used in data centres with a limited number of physical servers (ABACUS, no date).

1.3 Paravirtualization

Paravirtualization is similar to full virtualization as it uses the host-guest model, but the main difference is that, in paravirtualization, the guest operating systems are aware of each other and work together as a unified system (Broadcom, no date). This reduces the workload on the hypervisors, allowing efficient resource utilization. The operating system communicates with the bare-metal hypervisor using hypercall commands which enables them to create and manage the virtual machines (NUTANIX, 2023). However, instead of having the hypervisor directly on the hardware, it acts as a layer between the host's operating systems and the VMs. Currently, paravirtualization is not commonly used as servers now are designed to be able to support and work with bare-metal hypervisors without any limitations that came before. Some examples of paravirtualization are:

- Xen
- IBM LPAR
- Oracle VM for SPARC (LDOM)
- Oracle VM for X86 (OVM)

1.4 OS-level virtualization

OS-level virtualization, also known as containerization, does not need a hypervisor. Instead, it uses the host operating systems to create multiple virtual environments that share the host's resources (Broadcom, no date). Therefore, if the host's OS fails, the virtual servers will be affected. Each container uses the libraries and tools provided by the host OS, allowing them to operate independently. Since they share a single OS, additional components are not needed which reduces the overhead and improves resource efficiency. However, the containers must be using the same OS as the host. Examples of OS-level virtualization are:

- Oracle Solaris
- Linux LCX

- AIX WPAR

2.0 Using Hypervisor Technology to Address Challenges at MediCare Solutions Centre

MediCare Solutions is a well-known medical diagnostic center in Singapore. It specializes in radiology, pathology, and diagnostic imaging services for patients and healthcare providers. Over the years, it has collected an abundant amount of patient diagnostic data and medical imaging records. The tools that they used to store these records were different versions of medical imaging and diagnostic software. Moreover, since the number of patients is increasing and the center is expanding its services, they are in dire need of modernizing their infrastructure to ensure compatibility and efficiency in addressing these demands. Although, they are facing 2 challenges that hinder them from upgrading their current infrastructure. First, they are hesitant to upgrade because their old diagnostic software and patient data might not work with new systems. Second, they are uncertain about installing and testing new software on their main systems because of cybersecurity risks, especially concerns about protecting patient data and avoiding malware specific to healthcare.

2.1 Addressing how hypervisors help with Legacy System

Hypervisor technology enables multiple operating systems (OS) to run simultaneously on a single physical machine (Susnjara, 2024). This allows outdated operating systems to be replicated and run as virtual machines (Harwood, 2024). A hypervisor essentially creates and manages software-based versions of computers, enabling flexibility and resource efficiency (Harwood, 2024). Besides that, hypervisors allow hardware independence here virtual machines are separated from the hardware which enables the existing hardware to be updated without affecting the virtual machines running on it (Susnjara, 2024). Hence this will be beneficial if hardware components within the legacy system is outdated (Susnjara, 2024). Hypervisors also enable virtualization which separates a computer's hardware from its software here for the legacy system It allows older diagnostic software to run on modern hardware, avoiding the need to maintain outdated workstations (Denodo, 2024). Ultimately these reasons display how

virtualization and hypervisors technology address the legacy system challenges MediCare is facing.

2.2 Addressing How to Overcome Challenges in Installing and Testing New Software

Hypervisor technology can improve cybersecurity by isolating virtual machines from each other and the host, with this if one gets attacked with malware it won't spread to others (illumio, 2024). This helps protect sensitive patient data. Then, Hypervisors allows MediCare to run multiple VMs simultaneously which lets different software be tested compared to each other which improves decision-making. Furthermore, to test new medical imaging and diagnostic software, hypervisors can build isolated virtual machines. These sandboxes guarantee that patient data and the production environment are not jeopardised by possible malware or other problems (Greamo, 2011).

2.3 Factors to Consider for Choosing an Appropriate Hypervisor

There are two kinds of hypervisors Medicare could choose from, The First is type 1 hypervisor known as native or bare metal hypervisor which specializes in operating directly on the host hardware to oversee guest operating systems (Redhat, 2023). The second one is a type 2 hypervisor known as hosted hypervisor It functions by isolating guest operating systems from the host operating system, allowing virtual machines to run on the hardware layer (Red Hat, 2023). Furthermore, before opting for a specific hypervisor there are some factors MediCare should consider.

Hardware Compatibility

In Medicare Solutions it's crucial that the hypervisor can support existing data as it needs to directly communicate with the host machine's hardware to create and control virtual machines (Yfantis, 2020). By this MediCare can run older applications to access their data and ensure smooth integration with legacy systems.

Scalability

Since hypervisors allow multiple virtual machines to run on one physical machine (AWS,

2024), it can help to effectively scale with increasing number of patients and service expansion. A scalable hypervisor can handle increase in workload and support more services without needing extra physical hardware.

Security Features

Since MediCare solutions face challenges which as hesitance to upgrade their system due to cybersecurity threats and healthcare-targeted malware they need to choose a hypervisor that protects the software that runs the virtual machines (EDN, 2013).

Cost

The cost of hypervisor technology can be expensive as it comes with the hypervisor along with the software needed for scalability (Susnjara, 2024).

Overall Recommendation

MediCare solutions should use both types of hypervisors since both can attend to either one of the challenges they are facing. Type 1 Hypervisors for their testing environments and Type 2 Hypervisors for maintaining legacy software compatibility by using both hypervisors and use it for their future modernized infrastructure.

3.0 System Integration Challenges at GlobalEd Solutions

GlobalEd Solutions is one of the top providers of educational technology companies that develop an academic environment for colleges and institutions. However, the company is facing an issue regarding collaborative content development and distribution due to the inefficiency and incompatibility of integrating components made by geographically scattered teams. In order to address these issues, the company plans to utilise the virtualisation technology to implement a centralised platform for virtualized testing and development that is housed on their cloud infrastructure.



Incompatibility of contents and tools

The incompatibility of contents and tools is caused by diverse local environments. Each team uses various tools, platforms, development methodologies and runs on its own virtual machines which have different configurations and setups (Jacob Tenwick,2023). This uniformity has eventually led to integration issues in the final deliverable system.

Inefficiency of contents and tools

The inefficiency of content and tools is caused by inconsistent content development and testing by teams that are working independently in diverse locations. The lack of coordination and real-time collaboration has led to delays in feedback, misalignment, work replication and a longer integration process.

3.1 Utilization of Hypervisor Technology

GlobalEd Solution can effectively handle the challenges that the company is facing currently by implementing the hypervisor technology. This is because Hypervisor, an essential software layer that allows virtual machines (VMs) to run on a single host (Scale Computing, 2024) will create a unified virtualised environment for development teams that are working in a different location simultaneously without causing any integration hurdles. By implementing this technology, a centralised testing and development environment and

isolated virtual machine can be created to address the incompatibility and inefficiency of system integration.

Isolated Virtual Machines

The isolation of multiple virtual machines is often considered the most significant benefit that the hypervisor provides to address system integration issues. Each of the virtual machines will act like a fully functional, independent server with dedicated resources. This has made it possible for several people to work on the same server and ensure that confidential data will not be compromised (Charlotte Wright,n.d.). Other than that, the isolation can prevent the failure of one virtual machine when another virtual machine is making mistakes. Which means that any action made on one VM is exclusive to that VM. It ensures information can last permanently and will not cause any irreversible damage (Charlotte Wright,n.d.).

Centralised Testing Environment

The centralised testing environment is also one of the advantages that could help the company to overcome the system integration problem. The deployment of a uniform testing platform for all development teams ensures the consistency of the testing process. This ensures that the development teams can work in a standardized environment regardless of their geographical location. This will not only eliminate the integration issues but also speed up the development process.

Centralised Development Environment

Besides the centralised testing environment, the centralised development environment is also important to address the issue of system integration faced by GlonaEd Solution. By centralising the development environment, the development teams can use the same version of the programming language, tools and libraries, without the need to be in the same location. This has helped in eliminating the incompatibility issue. For example, the development team in Asia are working on the components of the video lecturer while another team in North America is working on the interactive simulations' components, they will be developed in the standardized platform in order to avoid any inconsistencies

throughout the integration stage.

Mechanism to Mitigate Delay and Accelerating Time-to-Market

The hypervisor technology provides a mechanism to streamline the integration process and prevent any errors that might arise during the testing and development process, which can help speed up time-to-market for the final system.

3.2 Additional benefit of hypervisor technology for GlobalEd Solutions

Cost Saving

The hypervisor technology often provides a more affordable service by consolidating several virtual machines on a single physical server, the expense of multiple physical servers can be eliminated. Hence, the cost can be cut down. Furthermore, the fewer physical server is used, the less power is necessary to operate these servers, and less space is required to be installed. This will result in the reduction of overall operational costs (Simran Dureja, 2024). GlobalEd Solution may be able to use these savings on other necessities.

Enhanced Scalability

Hypervisor technology allows GlobalEd Solution to scale its infrastructure up and down seamlessly and effectively based on its operational needs (Simran Dureja, 2024). For example, the company can scale their infrastructure up during the peak hour in order to handle large amounts of workloads, ensuring its functionality. In contrast, the resource might also scale down during off-peak hours to optimise the costs. The scalability of hypervisor technology allows the company to handle fluctuating workloads.

Improved Security

In hypervisor technology, each virtual machine is kept apart from others. This implies that if one VM is facing a security concern such as a malicious attack, the other VMs will still operate normally and remain unaffected. Furthermore,

Disaster Recovery

Hypervisor technology helps with the backing up and restoration of VMs by data replication. This is important for business continuity as well as minimising downtime of the system. Furthermore, a recovery plan is also essential to handle unexpected situations such as system failure, to ensure that the company regain the data and carries on with its operation.

4.0 GreenBites Cafe and Network Virtualization

GreenBites Cafe is a Malaysian eco-friendly cafe chain located in major cities such as Kuala Lumpur, and they have been facing challenges in managing their aging IT infrastructure. GreenBites cafe has been operating on aging servers created in 2017, they are prone to frequent malfunctions and poor performance due to the age and expired warranty. With limited financial resources and a small IT team, the company need to look for innovative solutions to centralize its IT infrastructure, and extend the lifespan of its hardware, and optimize costs. To address these problems, the company plans to transition to a virtualized network architecture. The primary goal is to centralize the management of its ordering system, customer information, and inventory tracking while reducing the needs of hardware replacement.

4.1 Role of Hypervisor

Hypervisor is a software that enables the creation of virtual machines (VMs). Hypervisors allow the creation and management of virtual machines while separating and isolating operating systems from the virtual machines. It facilitates server virtualization by allowing multiple operating systems to run separate applications on a single server while utilising the same physical hardware resources.

By implementing hypervisor, GreenBites cafe can address the limitations of their current IT setup. Hypervisors optimize the use of GreenBites existing server by enabling multiple VMs into a single server, thus maximizing the usage of their existing hardware and reducing the needs of newer hardware. This extends the lifespan of their aging server while allowing each virtual machine (VMs) to handle specific operational tasks, such as

ordering systems, customer data, and inventory tracking. By consolidating these functions, GreenBites cafe not only saves on hardware cost but also reduces their energy consumption. In case of failure, hypervisors can capture snapshots of the previous virtual machine, allowing GreenBites IT team to restore the VM and create backup or moving it to a different host, enabling quick data recovery.

In addition, centralized management within the hypervisors improves their IT operations. GreenBites IT team are able to manage and keep track of all VMs from a single interface, which simplifies management and improves efficiency. By adopting hypervisor infrastructure, GreenBites cafe ensures smooth operations across their outlets and online platforms and reduces downtime.

4.2 Centralized Control over Application and Desktop

Hypervisor allows centralized control over application and desktop by hosting them in the same systems. GreenBites cafe IT team and staff can have access to the same virtualized application and desktop across all locations and online platforms, ensuring seamless application. Updates across all platforms can be performed at the same time lowering IT workload and downtime. Centralized management also strengthens security by safeguarding sensitive data, better monitoring and reducing risk of unwanted attacks.

4.3 Maximizing IT Hardware Infrastructure Investment

With the implementation of hypervisor technology, GreenBites cafe may optimize resources and reduce the workload in physical hardware thus extending the lifespan of their aging server. Through virtualization, GreenBites cafe could combine several applications into a single server, reducing hardware, maintenance and power consumption. Furthermore, virtualization ensure long term sustainability by establishing an IT infrastructure that is adaptable to future business demand.

4.4 Benefits of Type-2 Hypervisor

Type-2 hypervisor is also known as a hosted hypervisor, runs on top of an existing operating system. They function as software programs inside the host OS and share with other applications, but with the capability of running and creating a virtual machine. Type-2 hypervisors are frequently used in desktop and development environments where workloads are less-resource intensive and operationally important, used in most cases when users only have one machine but need access to two or more operating systems at the same time.

Benefits of Type-2 hypervisor to GreenBites Cafe company:

- Ease of deployment: GreenBites IT team can easily deploy Type-2 hypervisor as they run on top of existing software.
- Efficient resource utilization: Type-2 hypervisor can help GreenBites cafe save hardware cost and maximize resource by consolidating multiple VMs into a single software.
- Energy efficient: Consolidation of multiple VMs reduces energy consumption thus aligning with GreenBite cafe dedication to sustainability.
- Cost effective: Suitable for GreenBites cafe limited financial resources as they do not require specialized hardware.
- Testing and development: For testing purposes, GreenBites cafe IT team can set up and isolate VMs for testing, experiments and looking for other solutions without disturbing their current systems.

5.0 Discuss how to persuade these two questions

Question1 : Discuss on the approach to convince your manager who wanted a less expensive, “good enough” virtualization solution to acquire a fuller-featured solution that would cost more.

In this issue, we can convince managers by emphasizing long-term value and return on

investment, emphasizing the function's support for corporate goals, and security and compliance.

Emphasizing Long-Term Value and Return on Investment

A more feature-rich virtualization solution, though higher in upfront costs, typically offers a comprehensive set of functionalities, including greater reliability, enhanced security, and superior scalability. For example, a fully equipped Type-1 bare-metal hypervisor can support larger workloads, deliver better performance, and provide stronger resource isolation, which is crucial for the company's future growth. For example, According to Gartner, as shown in the figure below, these technologies are expected to have a significant impact on business and society in the next 2 to 10 years (Perri, 2023).

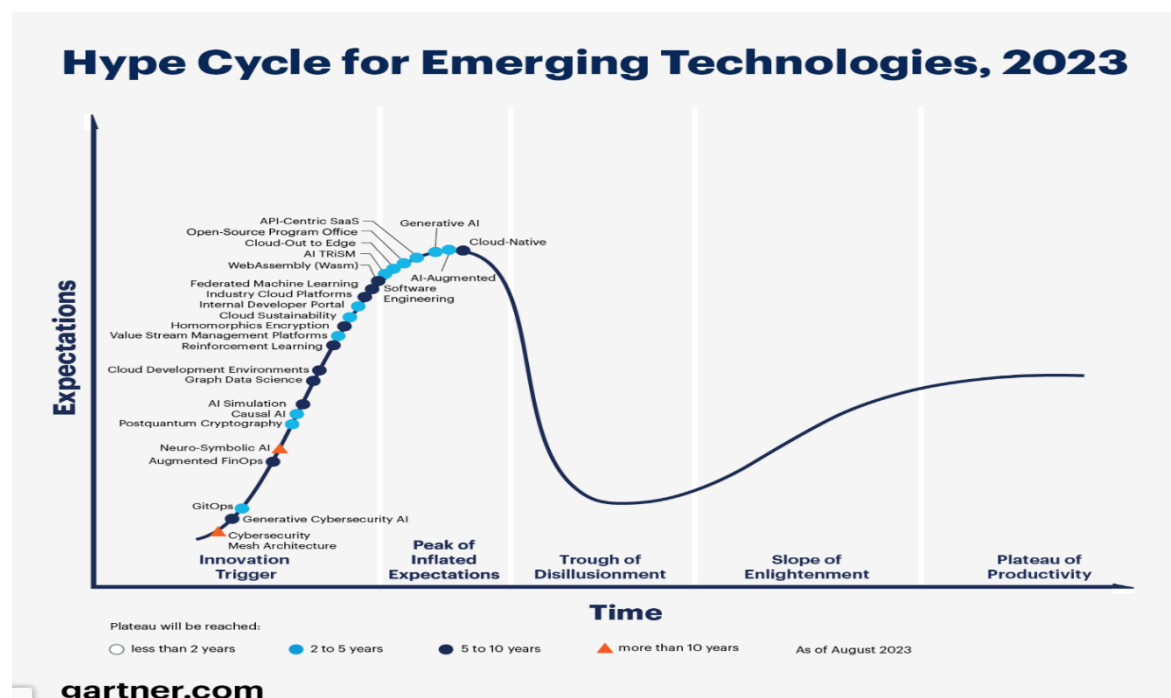


Figure 2- Hype Cycle for Emerging Technologies

Highlighting Support for Business Objectives

When a company's business needs high flexibility and scalability, a feature-rich virtualization solution can better support digital transformation goals. For instance, if a

company needs to deploy efficient virtualized data centres across multiple locations, advanced solutions offer multi-site support, automation tools, and disaster recovery capabilities that “good enough” solutions cannot achieve.

Security and Compliance

Modern enterprises face increasingly complex cybersecurity threats, making robust security features a critical requirement for virtualization solutions. A more advanced solution typically includes enhanced security mechanisms, such as virtual machine-based firewalls, real-time threat monitoring, and resource isolation capabilities. These are particularly crucial for industries like healthcare and finance, where data security is paramount.

Question2 : Discuss on the approach to convince your manager who wants to acquire a more expensive, fuller-featured virtualization solution to save some money and acquire a “good enough” solution.

In the second question, we can convince the manager by emphasizing the satisfaction of chip and core requirements, using a gradual expansion strategy, and reducing functional waste.

Managers by Emphasizing Long-Term Value and Return on Investment

When resources are limited, choosing a "sufficient" virtualization solution can effectively reduce enterprise operating costs. For example, Type-2 hosted virtualization solutions can run on existing operating systems, avoiding the investment in expensive hardware. This method is widely used in small and medium-sized enterprises, which greatly reduces deployment costs while meeting basic virtualization needs.

Proposing a Gradual Expansion Strategy

A “good enough” solution can be proposed as a transition strategy, and once the company budget allows or the requirements become clearer, it can be upgraded to a more feature-rich solution. This approach can meet short-term needs while leaving room for future scalability.

Minimizing Feature Waste

For businesses with well-defined requirements or slower growth rates, some functionalities of high-end virtualization solutions may remain unused. Choosing this solution prevents feature redundancy, ensuring that resources align with the company's actual needs.

References

Tardi, C. (2024). *What is moore's law and is it still true?* [online] Investopedia. Available at: <https://www.investopedia.com/terms/m/mooreslaw.asp>.

McClain , J. (2020). *Computer science: As Moore's Law slows down, the Insight Architecture Lab accelerates.* [online] William & Mary. Available at: <https://www.wm.edu/news/stories/2020/computer-science-as-moores-law-slows-down,-the-insight-architecture-lab-accelerates.php>.

Lyons, M. (2020). *How Moore's law drives hybrid cloud.* [online] Medium. Available at: <https://mikenovahunter.medium.com/how-moores-law-drives-hybrid-cloud-a6246465d790>.

Bigelow, S. (n.d.). *What is Server Virtualization?* [online] SearchITOperations. Available at: <https://www.techtarget.com/searchitoperations/definition/What-is-server-virtualization-The-ultimate-guide>.

Anon, (2020). *Three Types of Server Virtualization Explained - Abacus.* [online] Available at: <https://goabacus.com/three-types-of-server-virtualization-explained/>.

Amazon Web Services, Inc. (n.d.). *Type 1 vs Type 2 Hypervisors - Difference Between Hypervisor Types - AWS.* [online] Available at: <https://aws.amazon.com/compare/the-difference-between-type-1-and-type-2-hypervisors/>.

AWS (n.d.). *What is a Hypervisor? - Hypervisor Explained - AWS.* [online] Amazon Web Services, Inc. Available at: <https://aws.amazon.com/what-is/hypervisor/>.

Denodo. (2014). *Data Virtualization Overview.* [online] Available at: <https://www.denodo.com/en/data-management/data-virtualization>.

EDN (2013). *Embedded Systems Security - Part 3: Hypervisors and system virtualization*

- EDN. [online] EDN. Available at: <https://www.edn.com/embedded-systems-security-part-3-hypervisors-and-system-virtualization/> [Accessed 27 Dec. 2024].

Grearno, C. and Ghosh, A. (2011). Sandboxing and Virtualization: Modern Tools for Combating Malware. *IEEE Security & Privacy Magazine*, 9(2), pp.79–82. doi:<https://doi.org/10.1109/msp.2011.36>.

www.illumio.com. (n.d.). *Hypervisor - Cybersecurity 101 | Illumio*. [online] Available at: <https://www.illumio.com/cybersecurity-101/hypervisor>.

RedHat (2023). *What is a hypervisor?* [online] www.redhat.com. Available at: <https://www.redhat.com/en/topics/virtualization/what-is-a-hypervisor>.

IBM (2024). *Hypervisors*. [online] [ibm.com](https://www.ibm.com). Available at: <https://www.ibm.com/think/topics/hypervisors>.

Yfantis, V. (2020). *Hypervisor, what is it and its benefits*. | *Parallels Explains*. [online] Parallels Remote Application Server Blog - Application virtualization, mobility and VDI. Available at: <https://www.parallels.com/blogs/ras/hypervisor/>.

Microsoft (2024). *Hyper-V Technology Overview*. [online] [Microsoft.com](https://learn.microsoft.com). Available at: <https://learn.microsoft.com/en-us/windows-server/virtualization/hyper-v/hyper-v-overview>.

Tenwick, J. (2023). *Challenges of Working with Distributed Software Teams*. [online] RTeams. Available at: <https://rteams.com/challenges-of-working-with-distributed-software-teams/> [Accessed 27 Dec. 2024].

Scale Computing. (n.d.). *What is a Hypervisor?* [online] Available at: <https://www.scalecomputing.com/resources/what-is-a-hypervisor>.

Martin, R. (n.d.). *Not for general distribution. Intended for Advanced Sterilization Products Hypervisor Part 1-What is a Hypervisor and How Does it Work?* [online] Available at: <https://blackberry.qnx.com/content/dam/qnx/whitepapers/2017/what-is-a-hypervisor-and-how-does-it-work-pt1.pdf>.

Simran Dureja (2024). *What is a Hypervisor in Cloud Computing? Explore Here*. [online] Digitalregenesys.com. Available at: <https://www.digitalregenesys.com/blog/what-is-a-hypervisor-in-cloud-computing> [Accessed 27 Dec. 2024].

Macpherson, J. (2022). *What Is a Hypervisor? – Types, Benefits & How Does It Work?* [online] Park Place Technologies. Available at: <https://www.parkplacetechnologies.com/blog/what-is-hypervisor-types-benefits/>.

Bhardwaj, 1astha, Jain, R., Dhankhar, A., Sharma, V. and Com, A. (n.d.). *VIRTUALIZATION WITH HYPERVISOR*. [online] Available at: <https://ijcrt.org/papers/IJCRTICCC056.pdf>.

ResearchGate. (n.d.). (PDF) *VIRTUALIZATION: A REVIEW AND FUTURE DIRECTIONS Executive Overview*. [online] Available at: https://www.researchgate.net/publication/303854444_VIRTUALIZATION_A_REVIEW_AND_FUTURE_DIRECTIONS_Executive_Overview.

Gartner. (2023). *4 Exciting New Trends in the Gartner Emerging Technologies Hype Cycle*. [online] Available at: https://www.gartner.com/en/articles/what-s-new-in-the-2023-gartner-hype-cycle-for-emerging-technologies?utm_source=chatgpt.com.

Bigelow, S. and Kirsch, B. (2021). *What's the difference between Type 1 vs. Type 2 hypervisor?* [online] Techtarget. Available at: <https://www.techtarget.com/searchitoperations/tip/Whats-the-difference-between-Type-1-vs-Type-2-hypervisor>.

IBM (2023). *Hypervisors* | IBM. [online] [www.ibm.com](https://www.ibm.com/topics/hypervisors). Available at: <https://www.ibm.com/topics/hypervisors>.

XenServer. (n.d.). *Hypervisor: Definition, Types, and Benefits* | XenServer 8 Server Virtualization. [online] Available at: <https://www.xenserver.com/what-is-a-hypervisor>.

OVHcloud. (2020). *What is a Hypervisor?* [online] Available at: <https://www.ovhcloud.com/asia/learn/what-is-hypervisor/> [Accessed 27 Dec. 2024].

Amazon Web Services, Inc. (n.d.). *Type 1 vs Type 2 Hypervisors - Difference Between Hypervisor Types - AWS*. [online] Available at: <https://aws.amazon.com/compare/the-difference-between-type-1-and-type-2-hypervisors/>.

Systems, J.-J.B. (2023). *Understanding Type 2 Hypervisor: Enabling Efficient Virtualization in IT Solutions*. [online] Jaffer Business Systems (Private) Limited. Available at: <https://jbs.live/understanding-type-2-hypervisor/>.

Contribution of Students

Student Name	Contribution made:
1. Khor Yan Xin	3.0 System Integration Challenges at GlobalEd Solutions
2. Sureena Kulasingam	1.0 Relationship between Moore's Law and Virtualization
3. Heidi Victoria	3.0 System Integration Challenges at GlobalEd Solutions
4. Jyeshtha Yoshini A/P Jayakumar	2.0 Using Hypervisor Technology to Address Challenges at MediCare Solutions Centre
5. Wubin	5.0 Discuss how to persuade these two questions